

Windows Admin Center for Azure Local at Global Scale

A pragmatic, Microsoft-aligned WAC topology for a regulated pharma manufacturing estate running many Azure Local instances per Azure subscription, optimised for ARM-first operations, OT segmentation, and disconnected-site resilience.

Audience: Platform / Cloud / OT engineering

Sources: Microsoft Learn only

Scope: Dev/Test & Production

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Reviewers	<Add reviewer list>

1. Executive summary

Recommendation: a two-tier WAC topology.

- **Tier 1 — Regional HA WAC** (one HA pair per geo: e.g. US, UK, APAC). Active-passive failover-cluster install of Windows Admin Center, fronted by Microsoft Entra ID with Conditional Access & MFA. This is the daily-driver for the rare cases where ARM cannot perform the task.
- **Tier 2 — Per-site break-glass WAC**. A single Windows Server VM on the site's existing management VLAN (not on the Azure Local cluster, not in Azure). Powered off by default; activated only when WAN/ExpressRoute to the regional WAC is unavailable, or when an OT control mandates fully-local management.

This pattern preserves the stated objective of *as few WAC instances as possible* without trading away local-site resilience or OT isolation. It validates the "≈3 per region" hypothesis as **Tier 1 only**, with Tier 2 added as a documented break-glass capability.

Recommended

Two-tier (Tier 1 + Tier 2)

Few servers, full local resilience, ARM stays primary, OT-friendly.

Current state — not recommended going forward

WAC-in-Azure VM (today's pattern)

A WAC gateway running in an Azure VM *can* reach Azure Local nodes over ExpressRoute / S2S VPN, but it places the management plane on the wrong side of the WAN. See §3.1 for the full rationale.

Not recommended today

WAC in Azure portal (service)

Per Microsoft Learn, managing Azure Local cluster nodes via WAC in the Azure portal is documented as *preview*. Use only for opportunistic Arc-server tasks; do not design the operating model around it.

OT NOTE

This design assumes the OT control being invoked is "site-local management must remain available with no WAN dependency". If the actual OT control is different (e.g. "no inbound from corp at all", "no Entra ID at the site", "no domain trust"), the Tier 2 build changes. See §12 Open questions.

2. Terminology & gateway-mode disambiguation

Three distinct things are commonly called "gateway" in this space and have been mixed in prior threads. Lock these definitions down before any further design discussion:

Term	What it actually is	Where it lives
WAC gateway server	An installation mode of Windows Admin Center where the WAC <code>.exe</code> is installed on a Windows Server (or on a failover cluster for HA), and admins connect to it from a browser. This is what the HA deploy doc covers.	On-prem Windows Server (single VM or 2-node cluster).
Azure Arc gateway for Azure Local	An <i>Azure</i> resource that consolidates the outbound endpoints Azure Local nodes need to reach Azure. Reduces firewall allow-list size. Has nothing to do with WAC.	Azure (resource) + Arc proxy agent on each Azure Local node.
WAC in Azure portal	An Azure portal blade that lets you run a WAC-style UI against Arc-enabled servers / Azure Local nodes / Azure VMs without an inbound network path. Documented as <i>preview</i> for Azure Local cluster nodes.	Azure portal (no on-prem server install).

PLAIN ENGLISH

WAC gateway = the box you install. Arc gateway = an Azure-side traffic concentrator. They solve different problems and can coexist.

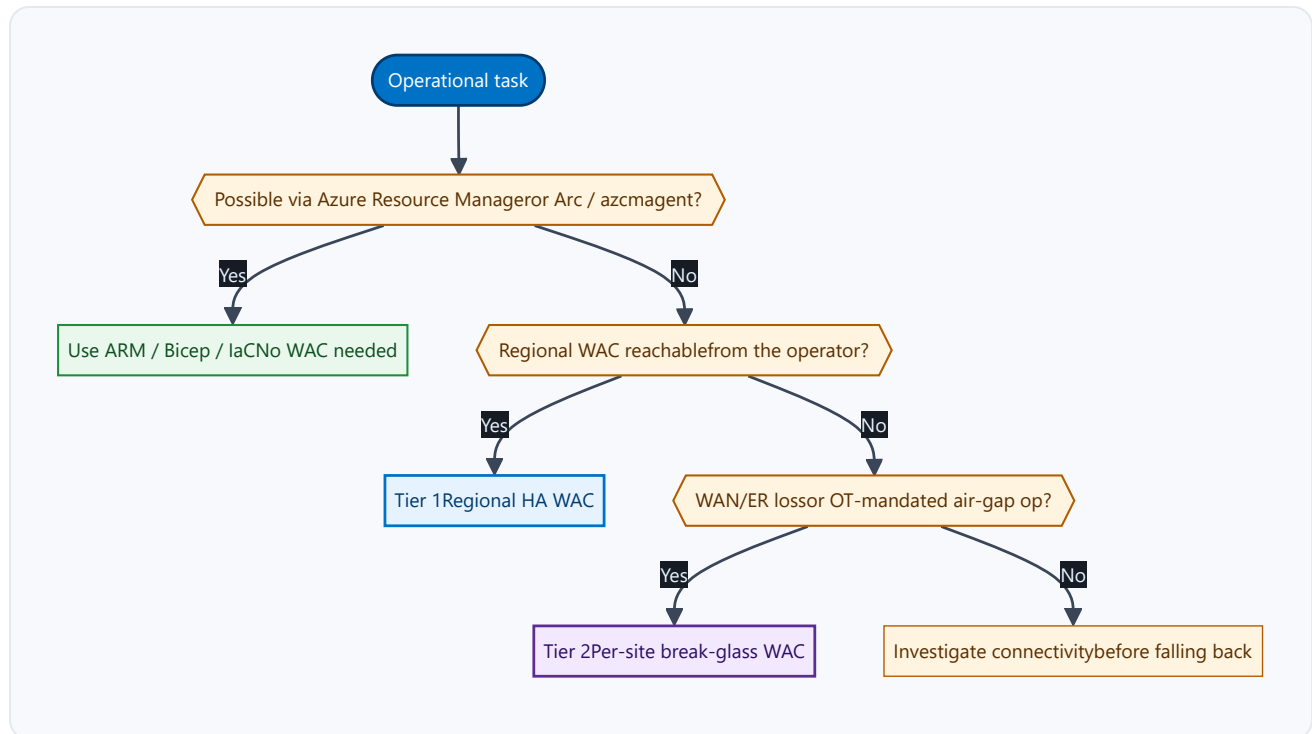
3. Topology decision matrix

Option	Local resilience (WAN loss)	OT segmentation fit	Server count (global)	Operational complexity	Verdict
1 WAC per Azure Local cluster	Excellent	Excellent	Highest (1× clusters)	Sprawl; many patch/cert lifecycles	Rejected — explicitly out of scope per the stated operating model; unnecessary given Tier 2.
Regional HA WAC only	None — site loses mgmt during WAN outage	Partial	Lowest (≈3 HA pairs)	Low	Insufficient for a regulated, manufacturing-floor estate.
Per-site HA WAC only	Excellent	Excellent	Very high (2× sites)	High	Overkill given ARM-first ops model.
Two-tier: Regional HA + per-site break-glass	Excellent (Tier 2 activates on WAN loss)	Excellent	≈3 HA pairs + 1 small VM per site (off by default)	Moderate	Recommended
WAC-in-Azure VM as primary (<i>current state</i>)	None — site loses mgmt during WAN/ER outage	Weak — mgmt traffic crosses corp WAN into the OT-adjacent VLAN	Low	Low day-to-day, high under failure	Not recommended — see §3.1.
WAC in Azure	None	Weak	Zero on-prem	Low	Preview for Azure Local — do not architect around.

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portal
(service)

Decision tree



3.1 Why WAC-in-Azure VM is not recommended for Azure Local at scale

This is the pattern currently in use. It works in normal weather; the problems are structural and surface during the exact failure modes a regulated manufacturing estate must plan for. The recommendation is to migrate Tier 1 to on-prem HA WAC and retain Azure VM access only as a transitional convenience.

Concern	Why it matters for Azure Local at a pharma site
1. WAN dependency for the management plane	The whole point of WAC for Azure Local is to do things ARM cannot — typically <i>during an incident</i> . If the incident is (or coincides with) ExpressRoute / S2S degradation, an Azure-hosted WAC is unreachable precisely when it's needed. On-prem Tier 1 + Tier 2 break-glass keeps management local to the failure domain.
2. Latency on interactive WinRM / PowerShell sessions	WAC operations are chatty PowerShell remoting + WMI calls between the gateway and the cluster nodes. Round-tripping every call from an Azure region back to a manufacturing site over ER adds tens to hundreds of ms per call; tools that issue many small calls (volume ops, cluster validate, live migration wizards) feel sluggish and time out under load.
3. Network direction is wrong for OT segmentation	Azure-hosted WAC must initiate inbound-equivalent flows (WinRM 5985/5986, SMB 445, RPC ranges) from Azure into the site's management VLAN — the segment adjacent to OT. On-prem WAC keeps all of that east-west inside the site. OT/network teams are far more comfortable approving the on-prem pattern, and it aligns with the spirit of ISA/IEC 62443 zone separation.
4. Single shared WAC across many regions doesn't scale cleanly	One Azure-hosted WAC managing US + UK + APAC clusters concentrates risk (one VM = one blast radius) and forces every operator session through one geography. The current operational pain you're seeing matches this pattern.
5. Authentication path complexity during Azure outages	Azure-hosted WAC typically authenticates operators via Entra ID. During an Entra ID regional incident, the on-prem nodes are still healthy but the management UI is unreachable. On-prem Tier 1 can also use Entra ID, but Tier 2 falls back cleanly to local AD / break-glass accounts.
6. Patching & cert lifecycle is no easier in Azure	The cited "easier to manage in Azure" benefit doesn't materially apply: WAC is still a Windows Server install with the same <code>.exe</code> , the same cert, and the same upgrade caveats (e.g. 2311 → 2410+ uninstall/reinstall) regardless of where the VM lives.

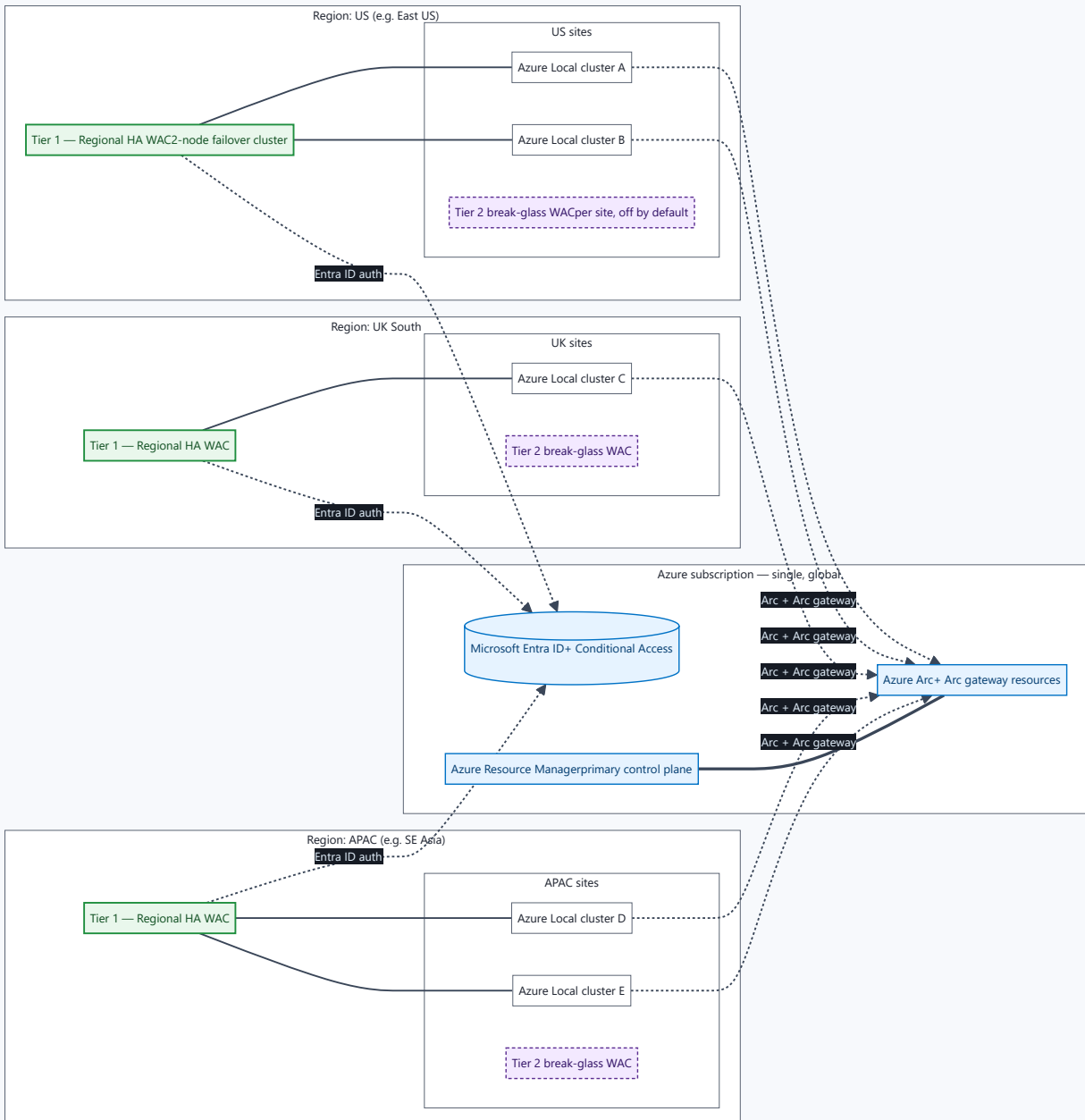
Concern	Why it matters for Azure Local at a pharma site
7. WAC-in-Azure-VM is not the same as "WAC in Azure portal"	The Azure portal experience for managing Azure Local cluster nodes is documented as <i>preview</i> and is feature-incomplete versus a full WAC gateway. Don't conflate the two when evaluating the future direction.

PRAGMATIC MIGRATION PATH

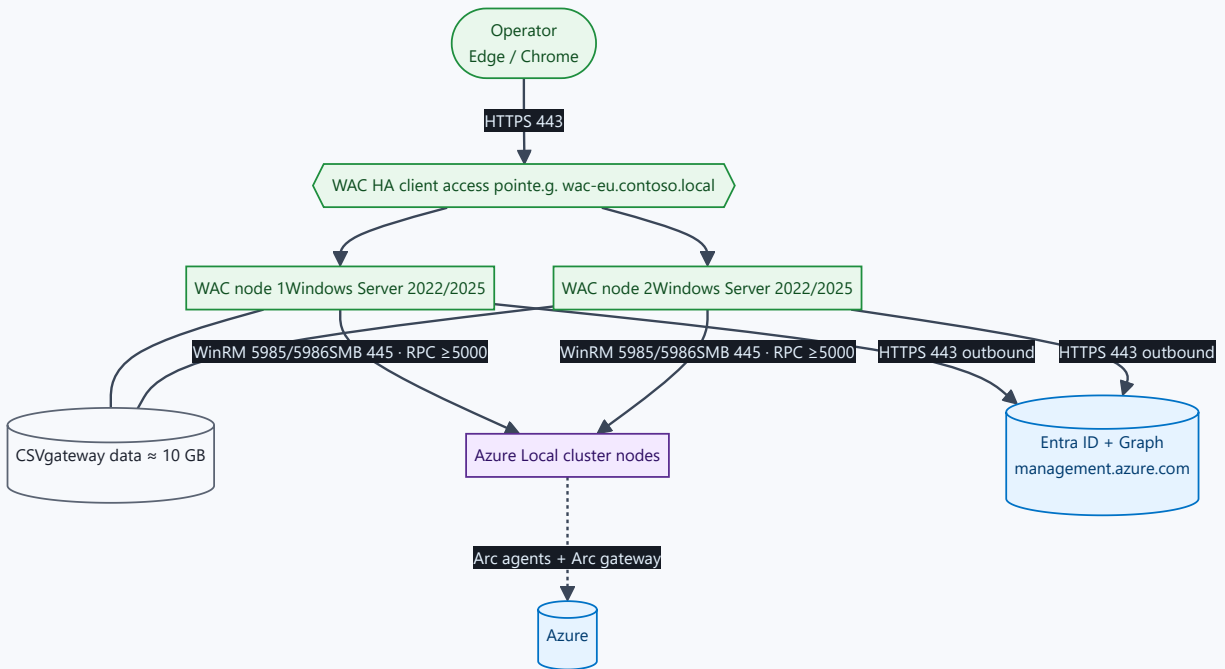
You don't need to delete the existing Azure WAC VM on day one. Suggested sequence: **(1)** stand up a Tier 1 HA WAC pair in one region as a pilot; **(2)** migrate that region's operators to it; **(3)** add Tier 2 break-glass at one site; **(4)** repeat per region; **(5)** retire the Azure WAC VM, or repurpose it as an admin jump host only.

4. Reference architecture

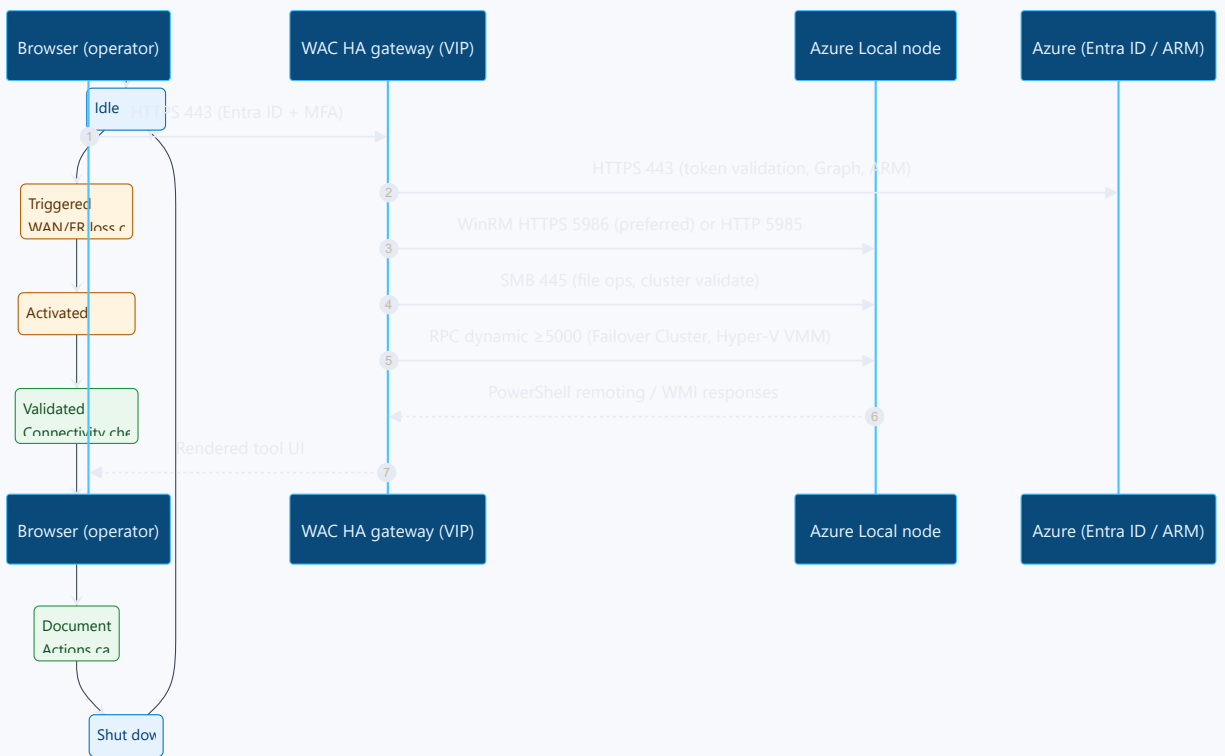
4.1 Global topology



4.2 Single-region detail



4.3 Port / protocol flow



5. Production pattern — Tier 1 HA WAC

5.1 Prerequisites

- 2-node Windows Server failover cluster (Windows Server 2022 or 2025 recommended).
- One Cluster Shared Volume (CSV), ~10 GB, for WAC persistent data.
- TLS certificate from a trusted CA, with the private key installed on *every* node; capture the thumbprint.
- WAC HA scripts: aka.ms/WACHAScript (`Deploy-GatewayV2Ha.zip`).
- WAC installer (`WindowsAdminCenter<version>.exe`) staged on a node.
- DNS A-record for the chosen client access point (e.g. `wac-eu.contoso.local`) and an optional static IP for the cluster generic service.

5.2 Sizing (starting point)

The HA install is **active-passive** per Microsoft Learn — one node serves traffic at a time and the other takes over on failure. This is not a load-sharing pair; size each node for full peak load.

Resource	Per node	Notes
vCPU	4 (suggested starting point)	Modernised gateway is multi-process / micro-service; scale up for many concurrent admins.
Memory	8–16 GB (suggested)	Increase if many extensions are loaded.
OS disk	127 GB (suggested)	Standard Windows Server image.
CSV (shared)	≥ 10 GB	Per Microsoft Learn HA prerequisites.
Network	1× mgmt NIC (+ optional cluster NIC)	Same VLAN as the Azure Local management network or routed to it.

Microsoft Learn does not publish prescriptive vCPU/RAM minimums for the modernized gateway; values above are pragmatic starting points for review.

5.3 Install the HA gateway

From a node in the cluster, with the script and installer locally:

```
$parameters = @{  
  ClusterStorage      = "C:\ClusterStorage\Volume1\Gateway"  
  ClientAccessPoint   = "wac-eu"           # becomes https://wac-eu.<domain>  
  StaticAddress       = "10.10.0.50"      # optional
```

```
InstallerPath          = "C:\Installers\WindowsAdminCenter2511.exe"
CertificateThumbprint = "AA11BB22CC33DD44EE55FF66AA77BB88CC99DD00"
}
.\Deploy-GatewayV2Ha.Deploy.ps1 @parameters
```

Validate and inspect with the sibling scripts (`Deploy-GatewayV2Ha.Validate.ps1` , `Deploy-GatewayV2Ha.Inspect.ps1`). To upgrade WAC later, re-run the same `Deploy-GatewayV2Ha.Deploy.ps1` with the new `-InstallerPath` .

UPGRADE CAVEAT

Direct upgrades from WAC HA **2311 and older to 2410 and newer are not supported** due to architectural changes in the modernized gateway. Plan an uninstall + reinstall when crossing that boundary. Source: Microsoft Learn (HA deploy guide).

5.4 Identity, RBAC & certificates

- **Identity provider:** Microsoft Entra ID for Tier 1, with Conditional Access + MFA + (optionally) PIM-eligible roles.
- **Gateway roles:** separate *gateway administrators* from *gateway users*.
- **Target-node RBAC:** use WAC's role-based access control (JEA-backed) so daily operators are not full local admins on Azure Local nodes.
- **Certificates:** use enterprise PKI; rotate via `-UpdateCertificate` on the same HA script. Avoid the `-generateSslCert` self-signed flag in production (60-day expiry).

6. Dev/Test pattern

One Windows Server VM (*not* a domain controller — installing WAC on a DC is unsupported). Local AD or Entra ID for auth. Self-signed cert acceptable for short-lived labs.

Build

1. Provision Windows Server 2022/2025 VM (4 vCPU / 8 GB).
2. Domain-join to the lab AD.
3. Download WAC installer from the Microsoft Evaluation Center.
4. Install in *gateway server* mode; bind to port 443.
5. Add lab admins to the *gateway administrators* group.

Parity vs. Production

- Same WAC tools and extensions are available.
- No HA / no failover — single point of failure (acceptable for lab).
- Use a *different* Entra ID app registration than production.
- Do not connect lab WAC to production Azure Local clusters.

7. Per-site break-glass WAC (Tier 2)

One small Windows Server VM per manufacturing site, on the site's management VLAN (the same network segment from which the local Azure Local instance was deployed). It is a **regular workload VM** — *not* a cluster node, *not* hosted in Azure — and is **powered off by default**.

7.1 Where it lives

Two acceptable hosting options, in order of preference:

1. **As an Azure Local VM on the site's own cluster** (recommended). It is a guest workload, not a cluster node. Survives WAN/ER loss because Azure Local continues to operate while disconnected from Azure (within the documented Arc grace period).
2. **On a dedicated small management host** at the site, if one already exists. Useful if the site standard requires the management plane to be physically separate from the workload cluster.

DO NOT

Do not host Tier 2 in Azure, on the Azure Local cluster *nodes* themselves, on a domain controller, or on the OT process network.

7.2 Build

- Windows Server 2022/2025, 2 vCPU / 4 GB / 127 GB. Domain-joined to the local site AD (or workgroup if AD is unavailable at the site).
- WAC installed in *gateway server* mode on port 443. Enterprise CA cert preferred; otherwise self-signed with documented thumbprint trust on the operator workstation.
- Local admin break-glass account documented in the site's existing privileged-access vault. Avoid hard dependency on Entra ID for the activation path.
- Recommended: keep the WAC version within one release of the regional Tier 1 instances to minimise behavioural drift between tiers.

7.3 Activation runbook

WHY NOT JUST LEAVE IT ON?

Leaving Tier 2 always-on doubles the attack surface per site and introduces drift between Tier 1 and Tier 2 RBAC. Powered-off-by-default keeps the OT segment quiet under normal operation and forces a deliberate, logged activation.

8. Network, firewall & OT requirements

8.1 WAC gateway → Azure Local cluster nodes (east-west, on-prem)

Source: Microsoft Learn — Firewall requirements for Azure Local.

Purpose	Source	Destination	Protocol	Port
SMB (file ops, cluster validate)	WAC	Azure Local nodes	TCP	445
WinRM HTTP (PowerShell remoting)	WAC	Azure Local nodes	TCP	5985
WinRM HTTPS (preferred)	WAC	Azure Local nodes	TCP	5986
RPC Endpoint Mapper	WAC	Azure Local nodes	TCP	135
Failover Cluster service	WAC	Azure Local nodes	TCP/UDP	3343
RPC dynamic (Failover Cluster, Hyper-V VMM)	WAC	Azure Local nodes	TCP	≥ 5000 (open ≥ 100 ports)
Hyper-V Live Migration	Mgmt system	Hyper-V hosts	TCP	6600
Hyper-V VM Management	Mgmt system	Hyper-V hosts	TCP	2179

TIP

If WAC is installed with the "Use WinRM over HTTPS only" option, allow only 5986 and skip 5985.

8.2 WAC gateway → Internet (north-south)

Source: Microsoft Learn — WAC network requirements. All HTTPS / TCP 443.

FQDN	Purpose	When required
<code>aka.ms</code>	Acquiring & maintaining WAC	Always
<code>download.microsoft.com</code>	Acquiring & maintaining WAC	Always
<code>pkgs.dev.azure.com</code>	Extension management	Always
<code>*.vsblob.visualstudio.com</code>	Extension management	Always
<code>login.microsoftonline.com</code>	Entra ID auth	When using Azure Hybrid Services (Public)
<code>graph.microsoft.com</code>	Microsoft Graph	When using Azure Hybrid Services (Public)
<code>graph.windows.net</code>	AAD Graph (legacy)	When using Azure Hybrid Services (Public)
<code>management.azure.com</code>	ARM	When using Azure Hybrid Services (Public)

8.3 Browser → WAC and browser → Azure

FQDN	Purpose	When required
<code>winadmincenterassets.blob.core.windows.net</code>	Extension management	Always
<code>js.monitor.azure.com</code>	Extension management	Always
<code>nuget.org</code>	Extension management	Always
<code>announcements.blob.core.windows.net</code>	Extension management	Always
<code>browser.events.data.microsoft.com</code>	Telemetry	Optional
<code>login.microsoftonline.com</code> , <code>graph.microsoft.com</code> , <code>graph.windows.net</code> , <code>portal.azure.com</code>	Azure Hybrid Services	When used

8.4 Arc gateway for Azure Local — orthogonal to WAC

The customer plans to use the **Azure Arc gateway** for Azure Local. This is unrelated to WAC topology but materially reduces the Azure-outbound endpoint list each Azure Local node needs. Key facts (Microsoft Learn):

- Enable Arc gateway **during deployment** of new Azure Local instances on version 2506+. It cannot be enabled post-deployment today.
- Arc gateway does **not** support TLS-terminating proxies.
- A short list of endpoints (e.g. `aka.ms` , `login.microsoftonline.com` , `management.azure.com` , `<region>.his.arc.azure.com` , `<arcgwid>.gw.arc.azure.com` , Key Vault, Cloud Witness, CRL/OCSP) is *not* redirected and must remain in the firewall allow-list.
- Azure Arc Private Link Scopes are not supported by Azure Local; Arc endpoints (`*.his.arc.azure.com` , `*.guestconfiguration.azure.com` , `*.dp.kubernetesconfiguration.azure.com`) must resolve to public IPs from the nodes / ARB / proxy.

HTTPS / TLS inspection is not supported for Azure Local outbound traffic. This includes Entra ID tenant restrictions v1. Coordinate with the OT/network team early — this is the single most common deployment blocker on regulated estates.

8.5 OT segmentation pattern (recommended)

- WAC gateway servers (Tier 1 and Tier 2) sit in the same management VLAN as the Azure Local cluster's management network — never on the OT process network and never on the storage / RDMA fabric.
- Operator browser traffic enters via a documented jump path (e.g. PAW or VDI) → Tier 1 WAC VIP only.
- Tier 2 has no inbound from corp WAN; only the local site PAW reaches it.
- All east-west WAC→node traffic stays inside the site's mgmt VLAN; only Azure-outbound traffic (443) traverses the site firewall, ideally via the Arc gateway.

9. Security baseline

Control	Setting
Identity provider (Tier 1)	Microsoft Entra ID + Conditional Access + MFA (smartcard optional via AD).
Identity provider (Tier 2)	Local AD / local groups; documented break-glass accounts in privileged vault.
Gateway access roles	Separate <i>gateway administrators</i> from <i>gateway users</i> .
Target-node RBAC	WAC RBAC (JEA endpoint) — Administrators / Readers / Hyper-V Administrators built-in roles.
Certificates	Enterprise CA, private key on every HA node, rotated via <code>- UpdateCertificate</code> .
Where WAC may NOT be installed	Domain controllers — unsupported per Microsoft Learn.
Local management of self	Don't use WAC to manage the same server it's installed on.
Auditing	WAC logging enabled; forward to SIEM (e.g. Microsoft Sentinel) via Windows Event Forwarding or AMA where supported.

10. Operations & lifecycle

- **Patching:** WAC non-preview versions are supported until 30 days after the next non-preview release. Stay within one release of current.
- **Version skew:** 2311 → 2410+ requires uninstall + reinstall on HA. Plan a maintenance window.
- **Cluster Manager extension:** ensure version $\geq$ **5.2.6** on every WAC instance before performing volume operations on Azure Local; otherwise update to WAC 2511 build 2.6.6.18 or higher (or use WAC-in-Azure-portal extension 0.76.0.0+). Source: WAC known-issues, April 2026 update.
- **Tier 2 drift control:** quarterly job to power on each break-glass VM, patch, validate, power off. Track in change management.
- **Backup:** WAC HA stores config on the CSV — back up the CSV and the cert. Tier 2 is stateless enough that re-imaging is the recovery path.

11. VM console / out-of-band access

WAC does **not** solve the "I need a console when WAC itself is broken" problem. Treat this as a separate, parallel capability and document it in the same runbook:

- OEM BMC (iDRAC / iLO / XClarity) on a dedicated, isolated mgmt VLAN, reachable only from a site PAW.
- Hardened jump host per site for serial-over-LAN / KVM access.
- Document MFA and credential rotation for BMC accounts; these are typically the most-overlooked privileged accounts in pharma OT estates.

12. Open questions for the customer

WE NEED ANSWERS TO FINALISE TIER 2

1. Which specific OT control is being invoked when you say "OT security requires X"? Is it ISA/IEC 62443 zone separation, a site-specific change-control rule, or a regulatory interpretation? The Tier 2 build differs for each.
2. Does each site have its own AD, or is AD homed regionally? This decides how Tier 2 authenticates with no WAN.
3. Confirm ExpressRoute / WAN design: single-circuit per site, or dual? Tier 2 only justifies its existence under realistic single-failure scenarios.
4. Confirm enterprise PKI source for WAC certificates and rotation cadence.
5. Confirm site count per region (drives the Tier 2 fleet size and patch rota).
6. Confirm whether Arc gateway will be enabled at *initial* Azure Local deployment (it cannot be added later).
7. Confirm operator workstation strategy (PAW vs. VDI vs. corp laptop) — affects browser-side firewall rules.

13. References (Microsoft Learn)

- Windows Admin Center overview
- What is Windows Admin Center?
- WAC — what type of installation is right for you?
- WAC — user access options
- Deploy WAC with high availability
- WAC — network requirements

- WAC — configure user access control
- WAC — known issues
- Azure Local — firewall requirements
- About Azure Arc gateway for Azure Local
- Register Azure Local machines with Azure Arc gateway
- WAC HA deployment script (Deploy-GatewayV2Ha.zip)
- Download Windows Admin Center (Evaluation Center)

All links resolved against Microsoft Learn at time of writing. No third-party or non-Microsoft sources are used.

How to read this document. Self-contained HTML — open in Edge or Chrome. `Ctrl+P` → Save as PDF produces a clean, paginated copy (the left navigation is hidden in print). Diagrams render client-side via the Mermaid CDN (`cdn.jsdelivr.net`); if your workstation blocks third-party CDNs, request the offline-bundled version, which inlines the Mermaid library so the file works fully air-gapped.